

CSE30301: Basic Math for AI - Midterm

Student ID :

Name :

Question 1 [30 points]

For each statement, mark **True (T)** or **False (F)**. You will receive +3 points for each correct answer, -2 points for each incorrect answer, and 0 points for unanswered questions.

- 1.1 If two random variables X and Y are independent, then $\text{Cov}(X, Y) = 0$. (T / F)
- 1.2 If $\text{Cov}(X, Y) = 0$, then X and Y are independent. (T / F)
- 1.3 Increasing the sample size reduces the variance of the sample mean. (T / F)
- 1.4 The p-value is the probability that the null hypothesis H_0 is true given the observed data. (T / F)
- 1.5 If A is invertible, then all singular values of A are non-zero. (T / F)
- 1.6 For any matrix A with full column rank, $A^T A$ is always invertible. (T / F)
- 1.7 Eigenvectors corresponding to distinct eigenvalues are orthogonal. (T / F)
- 1.8 The least squares solution always perfectly fits the data. (T / F)
- 1.9 The residual vector is orthogonal to the column space of X . (T / F)
- 1.10 Increasing λ in ridge regression reduces variance. (T / F)

Question 2 [20 points]

Let $X \sim \mathcal{N}(0, 1)$, and define $Y = aX + X^2$.

- 2.1 Compute $E[X]$, $E[Y]$, and $E[XY]$, showing all steps of your calculation. (9 points)

$$E[X] = \qquad E[Y] = \qquad E[XY] =$$

- 2.2 Compute $\text{Cov}(X, Y)$ as a function of a , showing all steps of your calculation. (11 points)

Question 3 [25 points]

You evaluate two algorithms, A and B, on the same task over 5 independent runs. The performance scores are as follows:

- Algorithm A: 11, 13, 15, 17, 19
- Algorithm B: 14, 16, 18, 20, 22

Answer the following questions.

3.1 Compute the sample mean of each algorithm. (5 points) $(\bar{X}_A = \quad, \bar{X}_B = \quad)$

3.2 Compute the sample standard deviation of each algorithm. (5 points) $(s_A = \sqrt{\quad}, s_B = \sqrt{\quad})$

3.3 Construct the 95% confidence interval for the mean performance of each algorithm. (For simplicity, you may use $z_{0.025} \simeq 2$ and $\sqrt{2} \simeq 1.41$.) (10 points)

$(A : (\quad) \pm (\quad), \quad B : (\quad) \pm (\quad))$

3.4 Can we conclude Algorithm B appears statistically better than Algorithm A on these confidence intervals? Briefly explain. (5 points)

Question 4 [30 points]

Let

$$A = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

4.1 Compute the eigenvalues of $A^T A$. (10 points)

4.2 Using the result from **4.1**, compute an SVD of A , i.e.,

$$A = U\Sigma V^T,$$

where the singular values in Σ are arranged in descending order. You must explicitly provide U , Σ , and V .
(15 points)

4.3 Using your SVD, compute A^{-1} . (5 points)

Question 5 [20 points]

Let

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad y = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

5.1 Compute QR decomposition $X = QR$, where:

- $Q \in \mathbb{R}^{3 \times 2}$ has orthonormal columns
- $R \in \mathbb{R}^{2 \times 2}$ is upper triangular

You must explicitly provide Q and R . (10 points)

5.2 Use your QR decomposition, solve the least squares problem:

$$\min_{\beta} \|X\beta - y\|.$$

You must clearly show how your solution is derived. (10 points)

Question 6 [25 points]

Consider polynomial regression of degree d :

$$\hat{f}(x) = \sum_{k=0}^d \beta_k x^k$$

- 6.1** Assume that $y = f(x) + \epsilon$, where $\mathbb{E}[\epsilon] = 0$ and $\text{Var}(\epsilon) = \sigma^2$. Let the expectation be taken over the randomness of the training data and noise. Show that: (15 points)

$$\mathbb{E}[(y - \hat{f}(x))^2] = \text{Bias}^2 + \text{Variance} + \sigma^2$$

- 6.2** What happens to bias and variance as d increases? (5 points)

- 6.3** How does regularization affect bias and variance? (5 points)

Question 7 [15 points]

Let X and Y be random variables. Prove that

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X|Y)] + \text{Var}(\mathbb{E}[X|Y]).$$

Question 8 [15 points]

Assume:

$$P(Y = 0) = P(Y = 1) = \frac{1}{2}.$$

Given:

- $\mathbb{E}[X|Y = 0] = 90, \quad \text{Var}(X|Y = 0) = 4$
- $\mathbb{E}[X|Y = 1] = 70, \quad \text{Var}(X|Y = 1) = 9$

8.1 Compute $\mathbb{E}[\text{Var}(X|Y)]$. (5 points)

8.2 Compute $\text{Var}(\mathbb{E}[X|Y])$. (5 points)

8.3 Compute $\text{Var}(X)$. (5 points)

Question 9 [20 points]

Let X_1 and X_2 be unbiased estimator of a parameter θ , with

$$\text{Var}(X_1) = \sigma_1^2, \text{Var}(X_2) = \sigma_2^2.$$

Assume X_1 and X_2 are independent, and define

$$X = wX_1 + (1 - w)X_2.$$

9.1 Compute $\text{Var}(X)$ as a function of w . (7 points)

9.2 Find the value of w that minimizes the variance. (8 points)

9.3 Interpret your result. Which estimator should receive a larger weight? (5 points)